

Leveraging AR-Driven Visual Storytelling to Enhance Communication of Complex Social Issues: Principles, Strategies, and Multifaceted Roles of Visuals

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ABSTRACT

By sharing principles and strategies for effectively using AR-driven Visual Storytelling to communicate complex social issues this paper adds to the fields of design and education. The insights offer visual designers a strong toolbox to create engaging experiences for audiences. Additionally, focusing on how to apply these strategies in various educational settings makes the proposed framework useful worldwide. By giving designers a clear way to blend augmented reality into visual storytelling, this study encourages creative communication about important social topics. This contribution not only boosts the impact of design but also helps education by promoting understanding across different cultures and sharing knowledge. Ultimately, the paper's approach connects design and education, opening new possibilities for teaching methods and positive changes in society.

CCS CONCEPTS

• Applied computing; • Education; • Collaborative learning;

KEYWORDS

Design Principles, Educational Strategies, Visual Design, Visual Communication, Visual Storytelling, Narrative Design, Engaging Experiences, Educational Settings, Augmented Reality Integration, Cross-cultural Understanding, Positive Educational Impact

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1 INTRODUCTION

Within contemporary academia, the integration of augmented reality (AR) technology into diverse fields, particularly visual communication, has sparked a transformative shift in creative expression [Papagiannis, 2017]. With more than 90% of information being absorbed non-verbally, as evidenced by different studies on speed and efficiency of visual processing in the brain [Medina, 2008],

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[Paivio and Csapo, 1973], [Schulze-Meeßen and Hamborg, 2023] the amalgamation of visual communication, augmented reality, and storytelling has emerged as a compelling avenue for crafting impactful visual narratives. Rooted in societal value, these narratives foster deep emotional connections, enhancing information retention and inspiring positive behavioural shifts.

The study delves into the intersection of design and societal influence, exploring how designers can effectively utilise AR technology to convey intricate social issues through visual narratives. The central challenge lies in uncovering effective strategies that form the foundation of AR-driven Visual Storytelling, a potent tool for addressing and reshaping complex societal challenges.

The primary aim of this study was to reveal how designers can leverage AR technology to prompt shifts in participation and behaviour, particularly regarding multifaceted social issues. By weaving AR capabilities into visual storytelling, designers can influentially reshape perspectives, evoke empathy, and stimulate positive actions, thereby contributing significantly to societal transformation.

2 GUIDING PRINCIPLES

Introduced is a set of critical outcomes with substantial implications, consisting of five guiding principles tailored to the unique challenges and opportunities within this field. The first principle, "Immersive Engagement," emphasises the importance of evoking emotional connections and deep engagement in the viewer through augmented reality. This, in turn, connects with the subsequent principles: "Narrative Structure," focusing on crafting narratives that effectively communicate complex social issues using storytelling techniques to resonate with the audience; "Empathetic Design," which promotes the development of empathy in design; "Positive Action," designed to prompt positive actions by leveraging AR capabilities to inspire viewers to take tangible, positive steps toward addressing multifaceted social issues; and "Cross-Cultural Adaptability," ensuring the adaptability of this approach across diverse cultural and educational contexts.

These principles draw upon insights from multiple disciplines, including design, psychology, and communication theories, ensuring a comprehensive and multidisciplinary foundation for crafting compelling narratives. These principles are carefully designed to empower designers to effectively convey intricate social issues through AR-driven Visual Storytelling and offer a structured pathway for crafting narratives that deeply resonate with audiences.

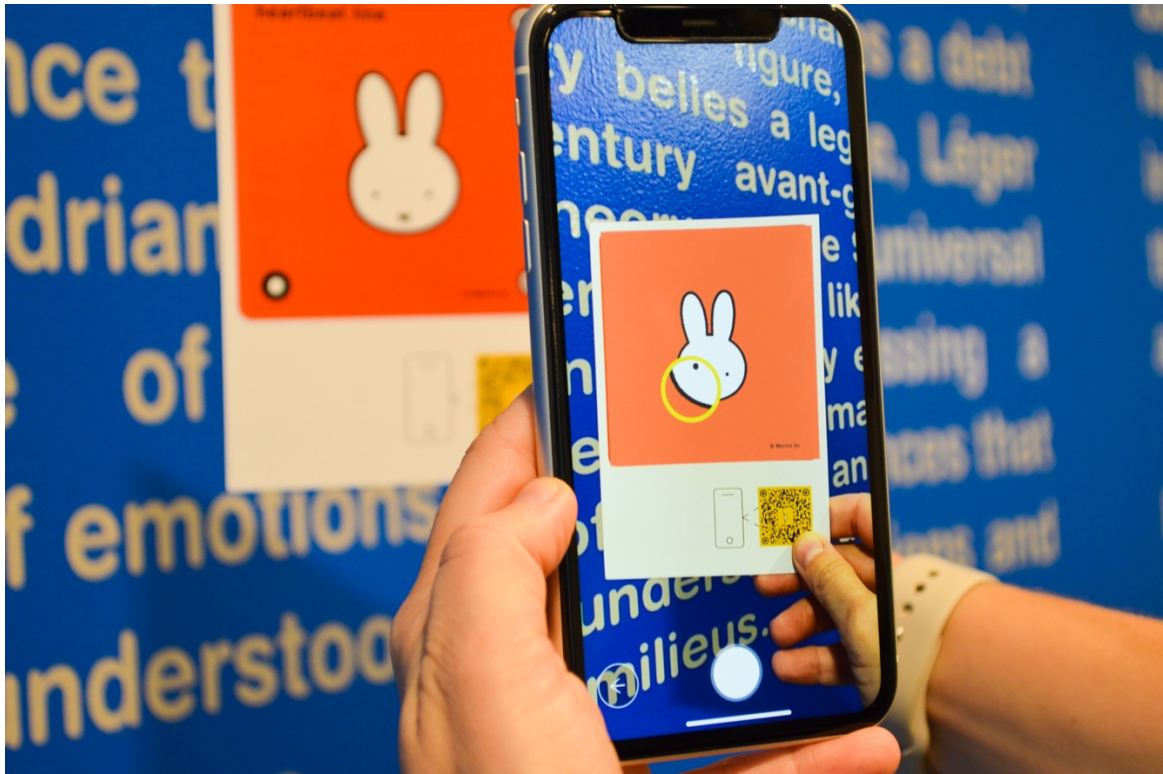


Figure 1: AR user test. Photograph by Anastasia Tyurina

3 APPLICATIONS IN THE BROADER CONTEXT OF EDUCATION

This paper emphasises the importance of placing these principles within the broader context of varied educational systems worldwide. Considering the global diversity of education structures, it is crucial to align the integration of innovative approaches with the prevailing norms. Design-driven education emphasises addressing complex, real-world issues, nurturing students' problem-solving skills and social consciousness. It fosters empathy and engagement, enabling students to consider the experiences and needs of others. This approach is adaptable and not bound by geographical or cultural constraints, making it effective in diverse educational settings. By combining design principles and educational practices, educators create learner-centric environments that stimulate creativity, critical thinking, and a sense of social responsibility, leading to positive changes in both teaching methods and society. This paper highlights that AR-driven Visual Storytelling has the capacity to align with and enhance educational paradigms across diverse cultures. This contextual approach enhances the nuanced understanding of the suggested principles and strategies, thereby facilitating the cross-cultural spread of knowledge and their practical implementation.

4 CONCLUSION

In summary, this work attempt to bridge the gap between design and education, offering opportunities for innovative teaching methods and positive societal changes. It achieves this by exploring the integration of augmented reality (AR) technology with visual communication and storytelling to address complex social issues. Through the application of AR-driven Visual Storytelling, the paper provides a framework with five guiding principles tailored for visual designers. These principles empower designers to effectively convey intricate social issues, drawing from multidisciplinary insights. Furthermore, the paper emphasises the importance of aligning these principles with various global education systems, making this approach adaptable and beneficial across diverse cultures. In essence, this work equips visual designers with the tools to engage audiences in transformative ways, ultimately strengthening the global applicability of the proposed framework.

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